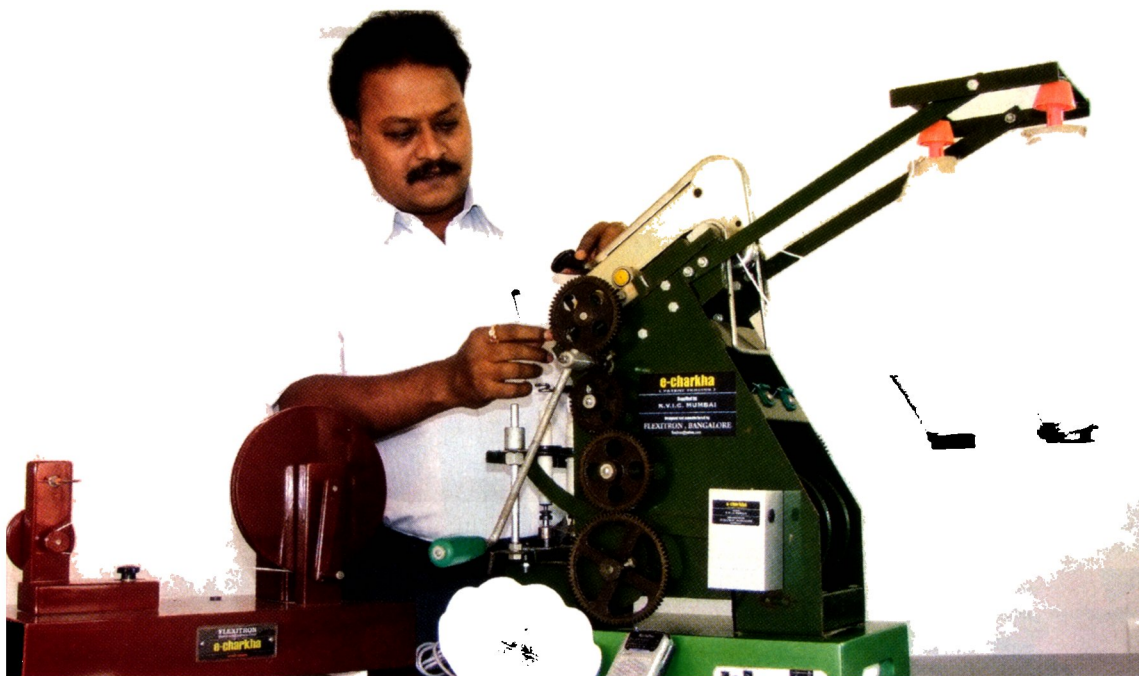


Spinning Power

An ingenious coupling of the age-old spinning wheel with an electrical dynamo, called the e-charkha, not only produces yarn but generates electricity as well.



HIREMATH WORKS ON THE E-CHARKHA

| By Nandini Vaish |

It's an invention Mahatma Gandhi would be proud of. Bangalore-based engineer R.S. Hiremath, 48, has developed a product that combines the fundamental principles of the age-old charkha with the power of a dynamo. The result: an electronic charkha which can light a bulb even as it spins fibre.

The e-charkha has a hand-driven spinning wheel that converts its rotational energy into electric energy. About 10 minutes of spinning can power a 25-watt bulb, enough to illuminate a 10-by-10-foot room for approximately an hour. The product is being tried in the rural areas of Myanmar, Bangladesh, Nepal and South Africa, where electricity is a scarce commodity. Andhra Pradesh is already planning to distribute these e-charkhas under the Funds for Regeneration of Traditional Industries scheme so that village homes can have individual lights. Even the Khadi and Village Industries Commission is sanctioning funds to agencies involved in khadi yarn production to buy the product and distribute it.

"I call it empowerment in a box," says Hiremath. Coming from a rural background, he remembers his

early days in Bijapur where there was no electricity. A young Hiremath was fascinated by his father's cycle and the light attached to it. While playing with his grandfather's charkha, he decided to mix the two. He first attempted that during his days at Gulbarga Engineering College, Karnataka, in the early 1980s, but the contraption turned out to be too bulky. After many trials and design changes, the light element was changed from a bulb to a light-emitting diode element. The cycle dynamo was replaced with a high-efficiency three-phase AC generator using rare earth magnets. Today, the e-charkha weighs about 10-12 kg and costs Rs 4,000 for a two-spindle unit and Rs 10,000 to Rs 12,000 for an eight-spindle unit. Hiremath has also attached a small transistor to his device, so that one can even listen to music while running it.

But that's not all from Hiremath. This compulsive inventor has also developed other products such as talking maps for the blind and an electric cycle rickshaw. In 2000, he was given the National Award for innovation in the disabled field for the talking maps. More power to Hiremath. ■